

Progress & Integration Report

AI ACADEMIC MENTOR

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
Instructor	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
Section	7C
Submission Date	19 August 2026
GitHub Repository Link	GitHub - Priyo-1158/cse4104-7c-t05-ai-academic-mentor: AI Academic Mentor - Personalized Learning Assistant for Students · GitHub

Team Information

Role	Member Name	Student ID	Contribution
Team Leader	Shadman Sadequeen Priyo	11230121158	Project coordination, GitHub management, integration testing
Frontend Developer	Musabbir Hossain Chayon	11230121168	React/Vite UI, API wiring, report authoring, deployment
Backend Developer	Sk. Nadirul Haque Nadir	11230121155	Express routes, MongoDB models, JWT authentication
AI Integration Lead	Shafin Kabir	11230121175	Gemini/HuggingFace routes, prompt design

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1. Complete Core Features

Per the assignment brief, every major feature promised in the proposal is listed below with an honest status, not an optimistic one. "In Progress" here specifically means: the code path is built and structurally correct, but is not currently producing the feature's actual intended output (a live Gemini response) because of an external configuration defect, not a missing feature.

Feature	Status	Notes / Reason	Target Date
User Registration / Login / Logout (JWT)	Completed	Real bcrypt + JWT auth confirmed in routes/auth.js and middleware/auth.js. No localStorage auth bypass remains — this was a real fix from the earlier flagged issue.	—
Role-Based Access (student vs admin)	Completed	Admin routes protected server-side via auth + adminOnly middleware, not just hidden in the UI. Confirmed in routes/admin.js.	—
Frontend-Backend CRUD (quiz / plan / summary / notes)	Completed	GET, POST, GET:id, PUT, DELETE all present and auth-protected on all four resources. Fully closes the "missing CRUD endpoints" gap flagged previously.	—
Progress / Analytics Dashboard	Completed	GET /api/progress and /api/progress/stats implemented and wired.	—
AI Chatbot	In Progress	Chat route, persistence to MongoDB, and frontend wiring all present — but every live call is silently failing over to fallback due to the invalid Gemini key.	Aug 25 — after key rotation
AI Quiz Generator	In Progress	Generation endpoint works structurally (prompt, JSON parse, fallback); blocked by the same Gemini key defect.	Aug 25 — after key rotation
AI Note Summarizer	In Progress	Same code path, same blocker.	Aug 25 — after key rotation
AI Study Planner	In Progress	Same code path, same blocker.	Aug 25 — after key rotation
CORS Configuration	Completed	Backend allowlist + wildcard *.netlify.app support confirmed in server.js.	—
Production Deployment Config	Not Started	Frontend still points at a literal placeholder backend URL ("YOUR-BACKEND.onrender.com"). Backend not yet confirmed deployed to Render/Railway.	Aug 24
Network / API Failure Handling	In Progress	Chat shows a fallback error message on request failure, but no request ever times out — there is no AbortController or axios timeout anywhere in the codebase. A hung connection freezes the UI indefinitely instead of failing gracefully.	Aug 25
Duplicate-Request Prevention	Not Started	Send button is disabled while a request is in flight, but the Enter-key handler in Chatbot.jsx calls sendMessage() directly with no guard — rapid Enter presses can still fire overlapping requests and duplicate DB writes.	Aug 24
AI-Specific Rate Limiting	Not Started	aiLimiter is fully defined in server.js but never mounted with app.use() — it is dead code. AI routes currently share only the generic 100 req / 15 min limiter.	Aug 24
Mobile / Responsive UI	Completed	Confirmed via submitted screenshots across chatbot, dashboard, planner and admin views at 425px.	—

2. Complete Frontend–Backend Integration

What is verified working

- Axios instance (frontend/src/services/api.js) attaches the JWT from localStorage to every request via an interceptor, and normalizes backend error shapes to err.message consistently across all pages.
- All CRUD resources (quiz, plan, summary, notes) round-trip through real MongoDB-backed Express routes, not mock data.
- Admin panel reads live stats/users from the backend, and unauthorized/non-admin access is rejected at the API layer, not just hidden in the UI.
- CORS is correctly scoped: local dev origins + CLIENT_URL + any *.netlify.app subdomain are allowed; everything else is rejected with a clear error.

3. Complete AI Integration

The core defect

backend/.env currently holds a GEMINI_API_KEY that starts with "AQ." — that is the shape of a short-lived Google OAuth access token (expires in roughly an hour), not a Gemini Developer API key, which must start with "Alza" and is generated at aistudio.google.com/app/apikey. This is not a guess: the code's own comment block at the top of routes/ai.js already documents this exact diagnosis, meaning the team identified the bug previously but has not yet rotated the key.

The practical effect: every Gemini call in processAIChat() throws, the code silently falls through to the HuggingFace fallback (Qwen 2.5-7B-Instruct), and if that also fails, to a hardcoded static text template. None of this is visible to a user in the UI beyond a "source" field in the API response — meaning any AI screenshot taken without checking that field may not show a real Gemini answer.

What is genuinely solid here

- The fallback chain itself (Gemini → HuggingFace → static) is a sound design decision and correctly implemented — this is good defensive engineering, not a flaw.
- Chat messages persist to MongoDB as Note documents when the user is authenticated, with source tagging so a developer can audit which provider actually answered.
- The migration off gemini-2.0-flash (deprecated June 2026) to gemini-2.5-flash is done correctly, with a documented lower-quality/higher-quota fallback model noted for rate-limit situations.

Fix required before this can be marked "Completed"

- Generate a real key at aistudio.google.com/app/apikey and replace GEMINI_API_KEY in .env. The startup warning already coded in ai.js will confirm the key format looks right — use it.

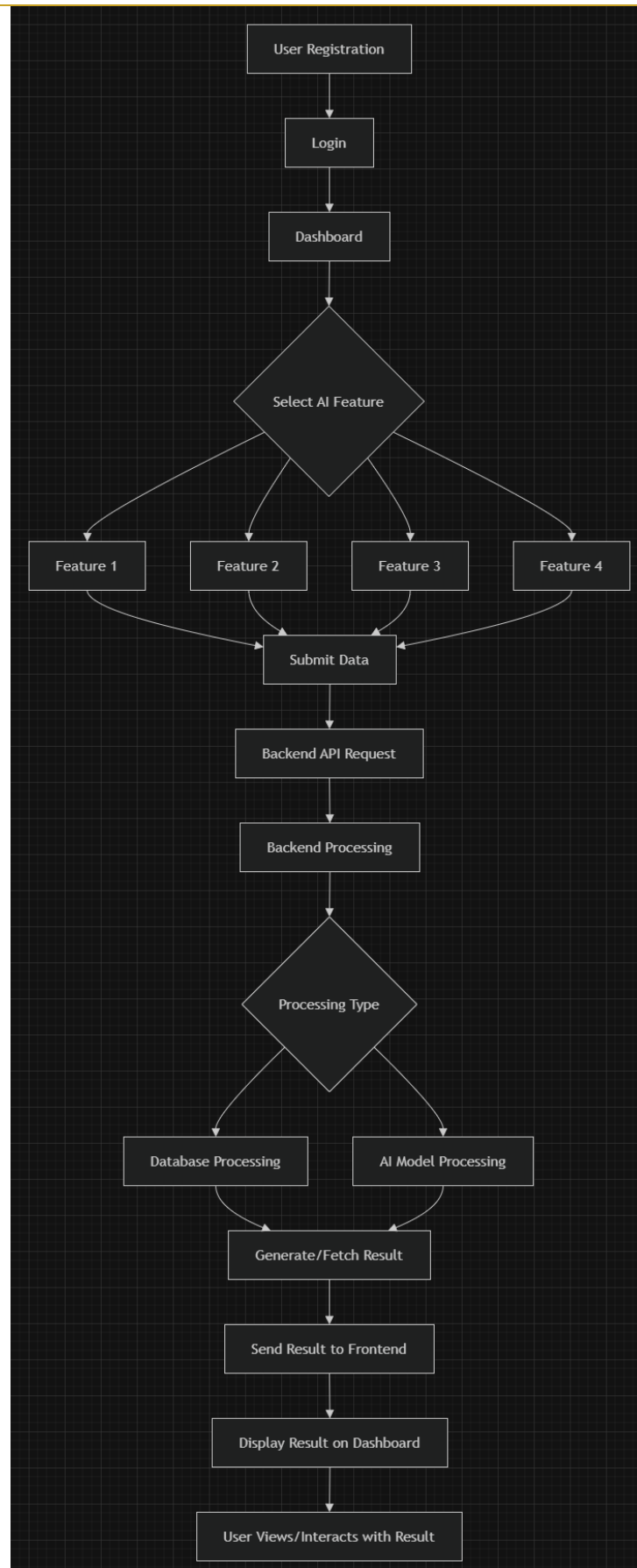
4. Complete User Authentication

This is the one area of the project that improved the most and is now genuinely solid. The previous flagged issue — a fake localStorage-based auth flow that bypassed the real backend entirely — is gone. Confirmed in the current codebase:

- Registration and Login: bcryptjs hashing on write, jsonwebtoken signing/verification on read — both real, both tested against the live MongoDB, not mocked.
- Logout: token is cleared client-side and the auth context resets; protected requests after logout are rejected by the backend, not just hidden in the UI.
- Session management: middleware/auth.js correctly distinguishes "no token" (401), "expired token" (401 with a specific message), and "invalid token" (401).
- Protected routes: ProtectedRoute.jsx blocks unauthenticated access client-side, and every sensitive backend route re-checks the token server-side — the frontend guard is a UX convenience, not the actual security boundary.
- Role-based access: admin-only endpoints require both a valid token and role === 'admin', enforced server-side in middleware/auth.js's adminOnly and applied via router.use(auth, adminOnly) in routes/admin.js — not just an AdminRoute component hiding a link on the frontend.
- ADMIN_ACCESS_CODE for admin self-registration is now a server-side .env gate, not a client-side JS check that anyone could read in devtools — this closes a real, previously-flagged security hole.

5. Complete Major User Workflows

The core end-to-end workflow — Registration → Login → Dashboard → Select Feature → Submit Data → Backend Processing → Database/AI Processing → Result Display — was traced through the actual code for each of the four AI features.



6. Improve UI/UX

Compared against the Week 05 approved design and the Week 07 baseline UI:

- Navigation: persistent sidebar with active-state highlighting across all 9 pages; consistent Dashboard → Feature → Back pattern.
- Loading / empty / success states: Study Planner and Quiz Generator both show a dedicated empty-state illustration with instructions before first use, and populated "Saved Plans" / "Quiz History" lists after.
- Error and success messaging: form-level inline errors on Register/Login (e.g. password mismatch) rather than raw browser alerts.
- Mobile responsiveness: verified at 425px across Dashboard, Chatbot, Quiz, Planner and Admin — sidebar collapses to a drawer, card grids stack to single-column.
- Visual consistency: dark navy theme with consistent card, button and badge styling maintained across every page, closing out the "broken layout" issues flagged in earlier UI reviews.

7. Handle Errors Properly

Each failure scenario listed in the assignment brief, tested directly against the current codebase:

Scenario (per assignment brief)	Current Behavior	Result	
Incorrect information entered	Register/Login show inline validation errors (e.g. password mismatch, invalid email).	Verified	
Required information missing	Backend returns 400 with a field-specific message; frontend surfaces it inline.	Verified	
An API call fails	Axios error interceptor normalizes to err.message; AI routes fall back through the Gemini→HuggingFace→static chain instead of erroring.	Verified	
Database is unavailable	Mongoose connection errors are caught at startup and logged; not yet tested against a live mid-session DB drop.	Untested	
AI service fails	Handled by design via the three-tier fallback chain — this is the strongest error-handling path in the app.	Verified	
Unauthorized user hits a protected feature	401 returned server-side regardless of frontend route guards; confirmed in middleware/auth.js.	Verified	
Network connection is interrupted	No AbortController or axios timeout exists anywhere in the codebase — a dropped connection hangs the UI indefinitely with no user-facing message.	Failing	

8. Review Project Scope

Comparing the current implementation against the Project Proposal, SRS, System Design, UI/UX Design, and AI Integration Plan:

What changed

- Frontend fully rebuilt from vanilla HTML/JS to React + Vite. Reason: maintainability and component reuse across the growing page count (9+ pages). This is a technology change, not a scope change — every planned screen still exists.
- Authentication rebuilt from a client-side localStorage mock to real JWT-backed auth against MongoDB. Reason: the mock was a security hole and didn't reflect a real system.
- Admin access-code check moved from client-side JS to a server-side .env gate. Reason: the original was trivially bypassable via browser devtools.
- AI model target changed from gemini-2.0-flash to gemini-2.5-flash. Reason: Google shut down 2.0-flash on June 1, 2026 — this was a forced, not optional, change.

What has not changed

- The original proposal's six core modules (AI Chatbot, Quiz Generator, Note Summarizer, Study Planner, Progress Dashboard, Smart Notifications) are all still in scope and structurally complete — nothing was cut.
- System design (MERN stack, JWT auth, MongoDB schema) matches the originally approved architecture.

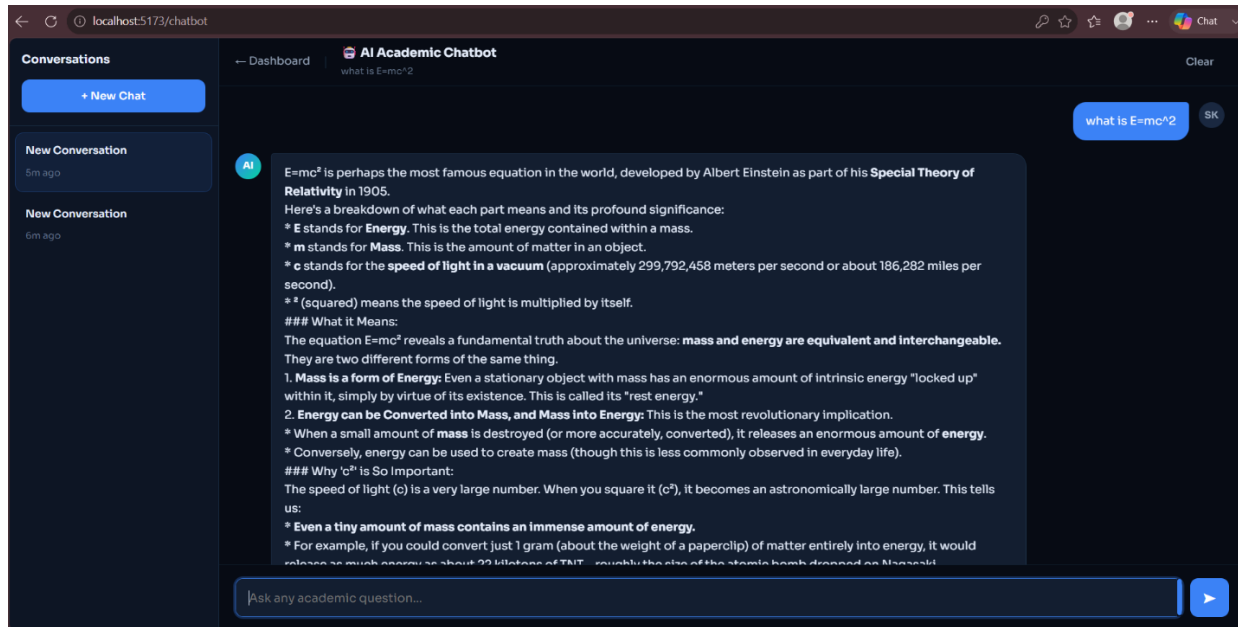
9. Major Problems Encountered & Solutions

Problem	Solution Implemented / Planned	Severity
Gemini calls fail 100% of the time — API key is OAuth-token-shaped, not a Developer key.	Regenerate a real key at aistudio.google.com/app/apikey ; paste into .env; restart server and confirm the startup log no longer warns about key format.	Critical
Deployed frontend cannot reach the backend — PRODUCTION_API_BASE is a literal placeholder.	Deploy backend (Render/Railway); set VITE_API_BASE as a Netlify build environment variable; redeploy and hard-refresh to confirm.	Critical
No request timeout anywhere — a dead connection hangs the UI forever instead of failing gracefully.	Add a global timeout to the axios instance in api.js and/or wrap fetch calls with AbortController + setTimeout.	High
Chat can fire duplicate/overlapping requests via the Enter key even while sending is true.	Add an `if (sending) return;` guard at the top of sendMessage() in Chatbot.jsx.	Medium
aiLimiter is defined but never mounted — AI routes have no dedicated rate limit.	Add `app.use('/api/ai', aiLimiter);` before mounting the AI router in server.js.	Medium
Partial Gemini key is logged to the console on every chat call.	Delete the debug console.log lines in processAIChat() before final submission.	Low

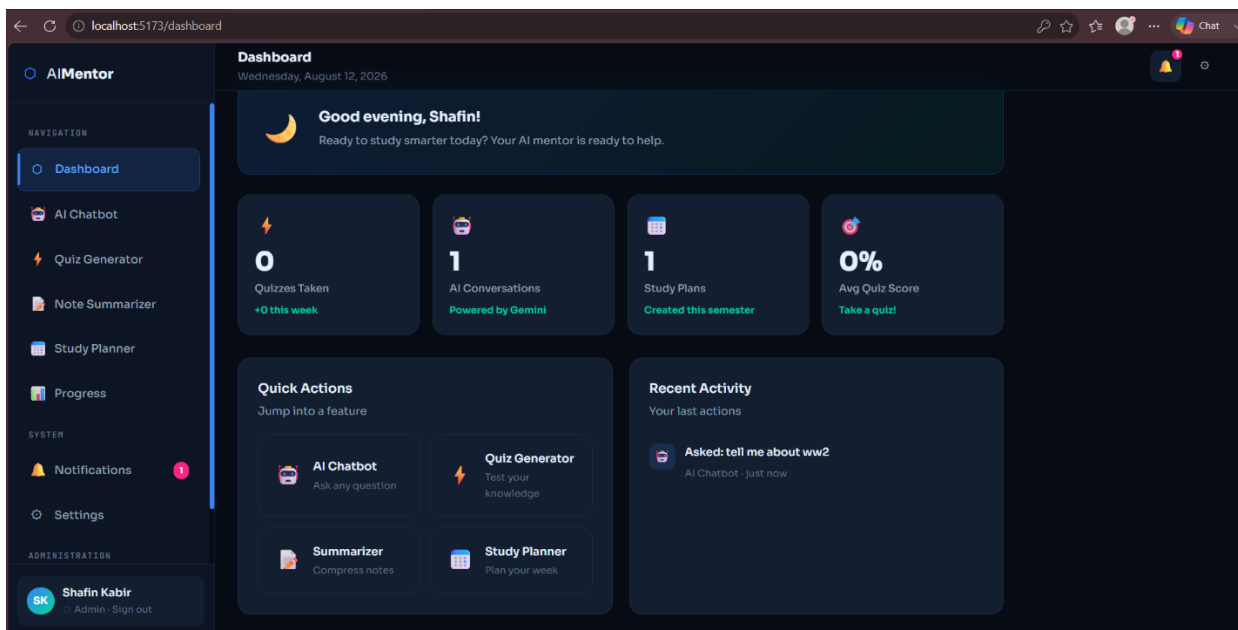
10. Current Project Screenshots

Selected from the team's full UI/UX screenshot set. Full desktop and 425px mobile captures for every page are maintained separately in the project's screenshots/ folder per the recommended repository structure.

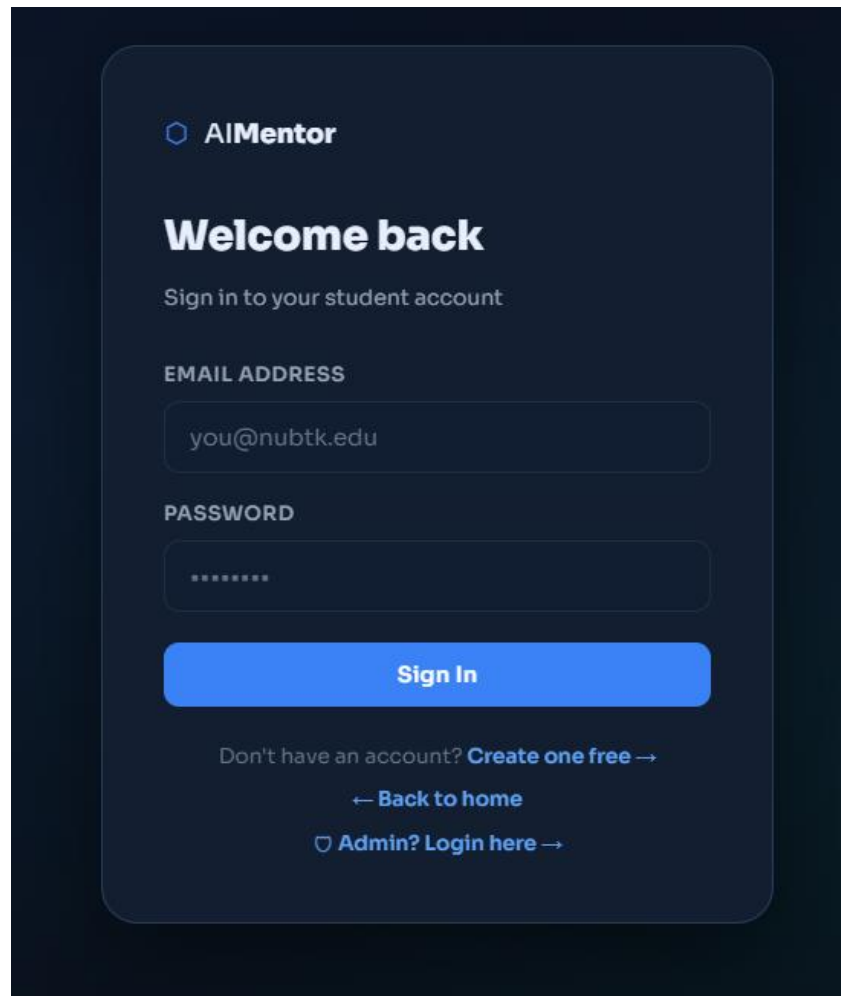
AI Chatbot:



Dashboard:

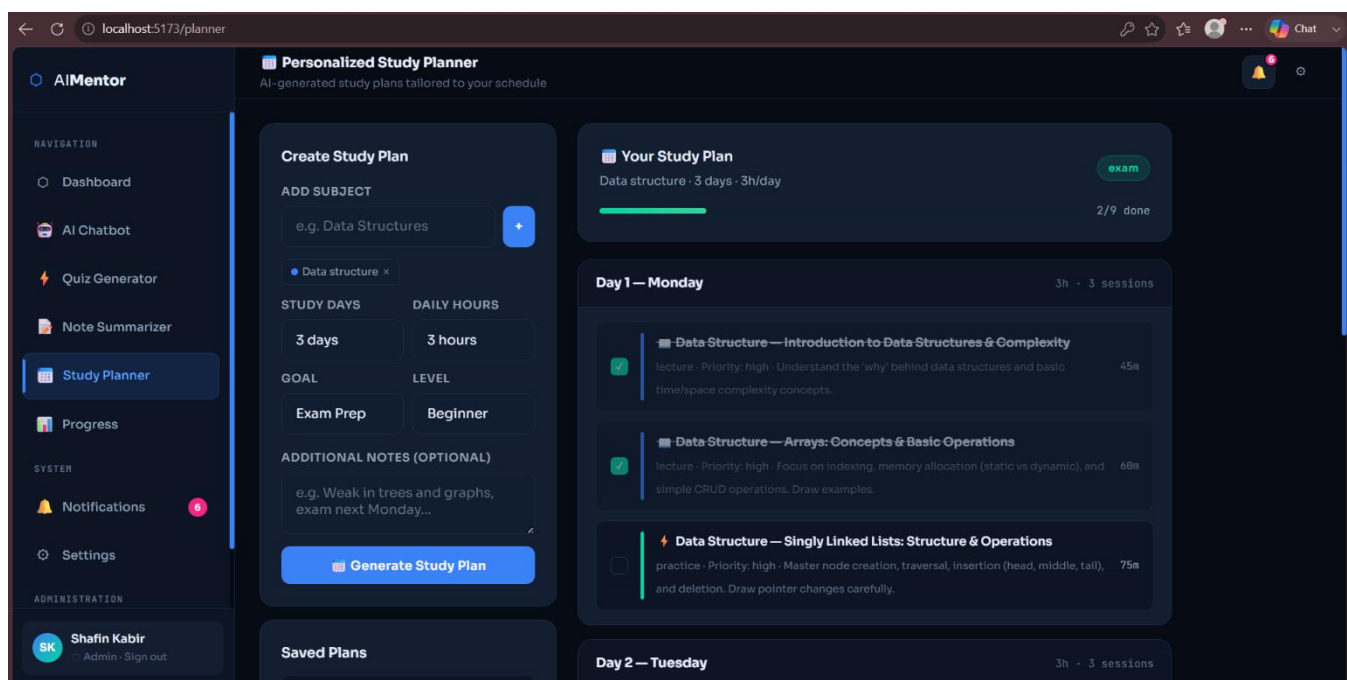


Login Page:



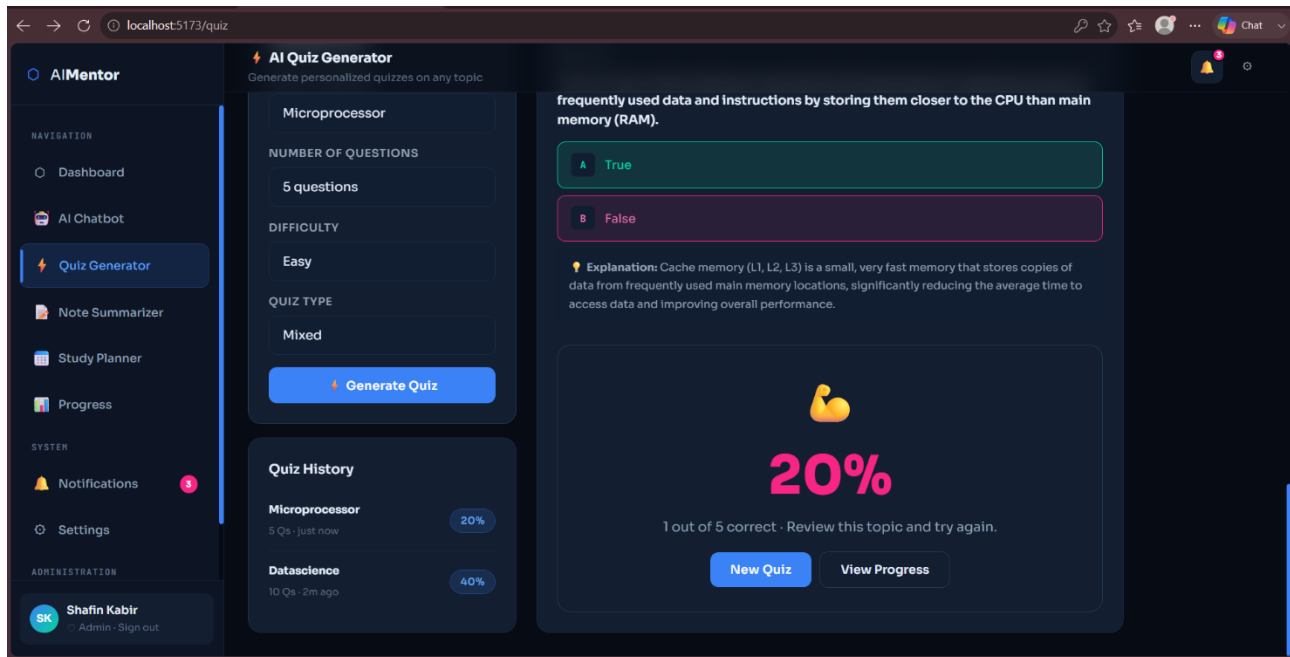
The login page for AI Mentor features a dark blue background with a central white card. At the top, the AI Mentor logo is displayed. Below it, the text "Welcome back" is prominently shown, followed by "Sign in to your student account". The form includes fields for "EMAIL ADDRESS" (with the placeholder "you@nubtk.edu") and "PASSWORD" (with masked characters). A large blue "Sign In" button is positioned below the password field. At the bottom of the card, there are links for "Don't have an account? Create one free →", "← Back to home", and "Admin? Login here →".

Study Planner:



The Study Planner interface is a dark-themed dashboard. On the left, a sidebar contains navigation links: Dashboard, AI Chatbot, Quiz Generator, Note Summarizer, Study Planner (highlighted), Progress, Notifications, and Settings. The main area is titled "Personalized Study Planner" and "AI-generated study plans tailored to your schedule". It features a "Create Study Plan" section with input fields for "ADD SUBJECT" (e.g., Data Structures), "STUDY DAYS" (3 days), "DAILY HOURS" (3 hours), "GOAL" (Exam Prep), and "LEVEL" (Beginner). Below these is an "ADDITIONAL NOTES (OPTIONAL)" field. A "Generate Study Plan" button is at the bottom. To the right, the "Your Study Plan" section shows a progress bar for "Data structure - 3 days - 3h/day" with a "2/9 done" status. It lists tasks for "Day 1 — Monday" (3h, 3 sessions): "Data Structure — Introduction to Data Structures & Complexity" (45m), "Data Structure — Arrays: Concepts & Basic Operations" (68m), and "Data Structure — Singly Linked Lists: Structure & Operations" (75m). A "Day 2 — Tuesday" section is partially visible at the bottom.

Quiz Generator:



Registration Page:

AI Mentor

Create account

Join AI Academic Mentor — it's free

FULL NAME

Your full name

EMAIL ADDRESS

you@nubtk.edu

STUDENT ID

1123012XXXX

DEPARTMENT

CSE

PASSWORD

Min 6 characters

CONFIRM PASSWORD

Repeat password

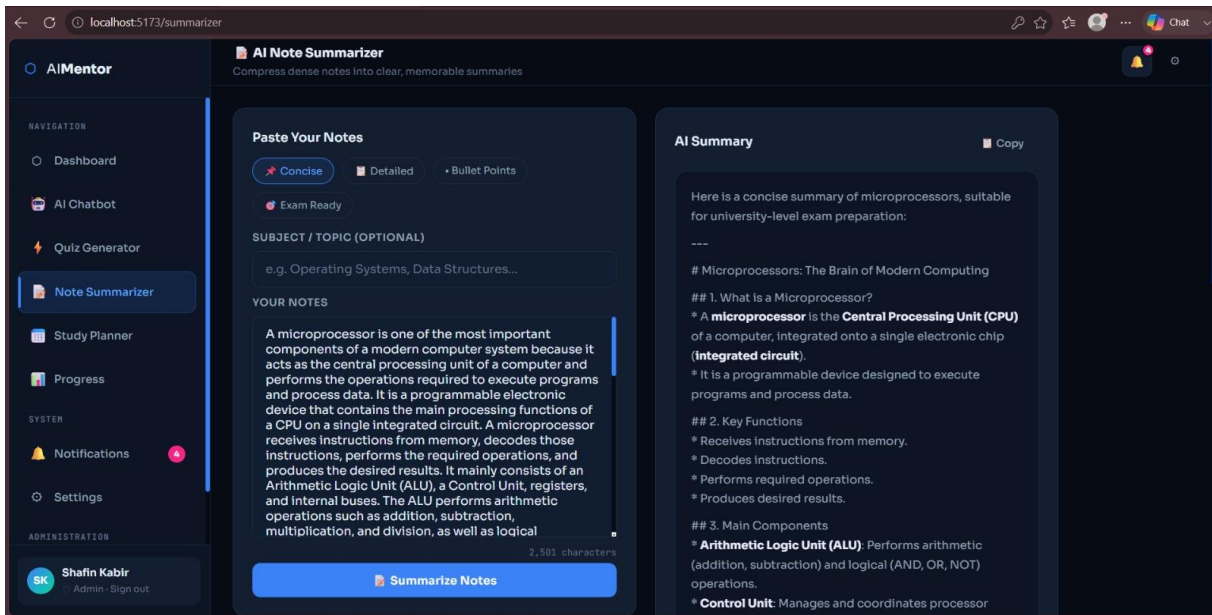
☐ Registering as Administrator

Create Account

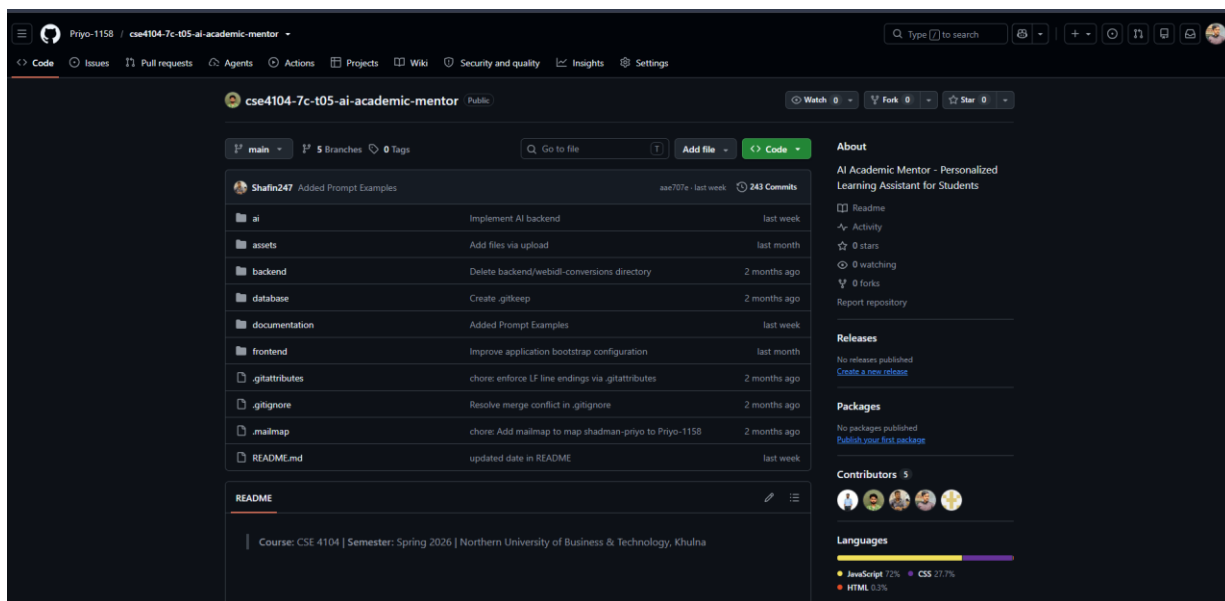
Already have an account? [Sign in](#) →

← [Back to home](#)

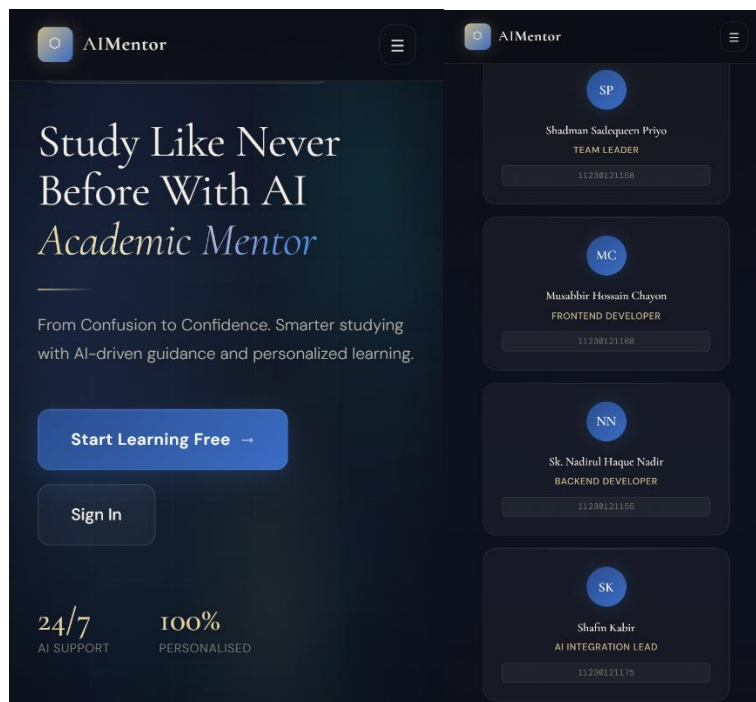
Note Summerizer:



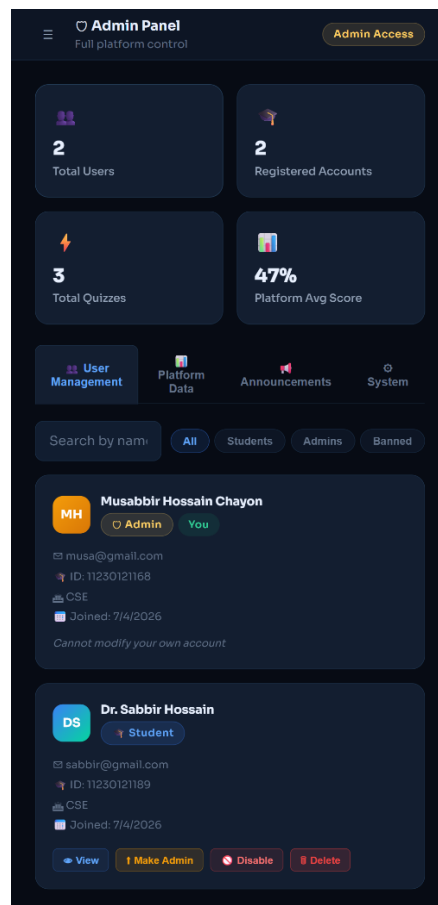
GitHub Repository:



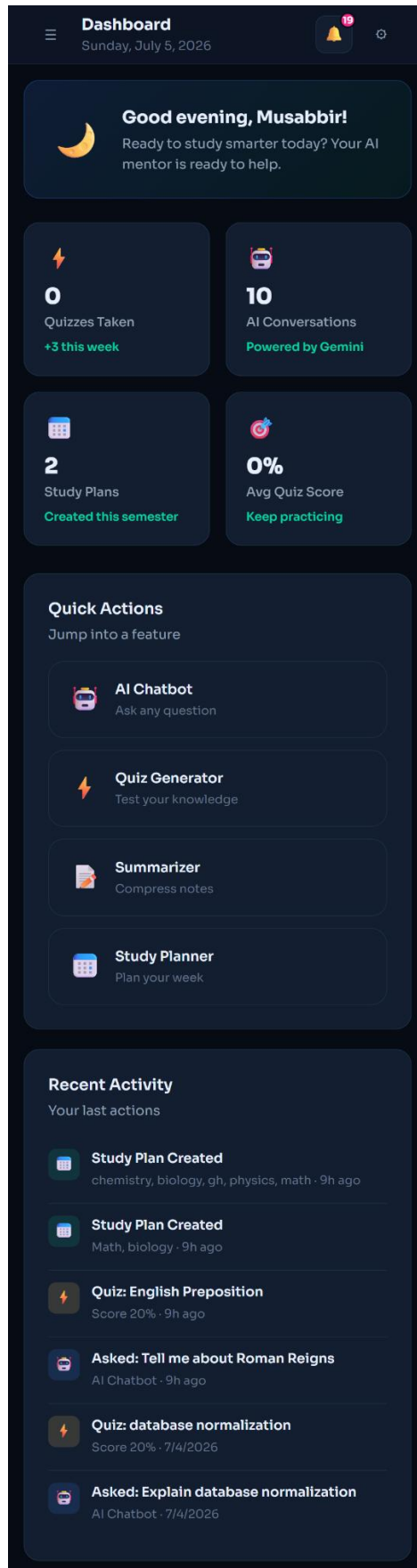
MobileView_Frontend_pages__index:



MobileView_frontend_pages_admin:



MobileView_frontend_pages_dashboard:



MobileView_frontend_pages_settings:

⚙ Settings

Manage your account and preferences

 Profile

 Data Management

 About

Profile Settings

Your account information

Name

Musabbir Hossain Chayon

Email

musa@gmail.com

Student ID

11230121168

Department

Not set

Role

admin

Joined

Unknown

Sign Out